

Inertia as a Spectral Separator: Geometry, Placement, and Controller–Network Hybridization in Low-Inertia Power Networks

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Abstract—Low-inertia studies usually treat inertia as an aggregate energy or frequency-security resource. This paper develops a complementary interpretation: inertia is also a *spectral separator* between network-dominated and converter-control modes, and its spatial distribution determines how strongly a critical closed-loop mode overlaps the controller spectrum. Starting from a controller-aware reduced nonlinear eigenvalue problem, we formulate a grid/controller separation margin through the Sylvester operator, prove a subspace-mixing bound, derive the $\eta^{-1/2}$ modal migration law under inertia scaling, and obtain a two-mode hybridization law near controller/network encounters. We then introduce a signed *inertia steering field* that maps equal additions of inertia at different buses into different directions of critical-pole motion. The theory is stress-tested on an AC-consistent reduced IEEE 39-bus SG/GFL model using 2,232 inertia/controller operating points and 1,200 equal-total-inertia placements. With the synchronous-inertia budget fixed at 30% of nominal, the placements span spectral abscissas from -0.326 to $+1.151 \text{ s}^{-1}$; only 316 are stable. Stable placements allocate, on average, 22.2% of the fixed budget to bus 39 versus 8.25% for unstable placements, and the highest placement-entropy quintile is stable 37.5% of the time versus 15.4% in the lowest quintile. A local inertia sweep at bus 31 exhibits a re-entrant stable–unstable–stable window with crossings at 0.1507 and 1.5169 of nominal bus-31 inertia. At a uniform 25% inertia level, identical effective-inertia increments applied at different synchronous buses move the critical pole in opposite directions; bus 33 stabilizes the case whereas buses 32, 35, and 36 move the mode further right. The minimum added effective inertia required for a first stability crossing ranges from 0.826 at bus 33 to 55.57 at bus 35, while three buses do not cross within a search range exceeding 593 effective-inertia units. By contrast, a 4.86% PLL proportional-gain retuning restores stability without replacing the missing inertia. These results support a planning principle: low-inertia adequacy is partly a *spectral-placement and controller-detuning problem*, not only an inertia-quantity problem.

Index Terms—Low-inertia power systems, inverter-dominated grids, virtual inertia, phase-locked loop, modal interaction, spectral separation, Sylvester equation, grid-following inverter, model reduction, controller-network interaction.

I. INTRODUCTION

LOW-inertia power systems are commonly analyzed through frequency nadir, RoCoF, equivalent inertia, damping, and small-signal eigenvalues. These quantities remain essential, but converter-rich systems add a second difficulty: their controller dynamics occupy time scales that increasingly

overlap with network and electromechanical dynamics. Recent work has documented increased model stiffness in low-inertia systems and the resulting difficulty of preserving a clean slow/fast decomposition [1]. At the device level, PLL and virtual-inertia loops can interact adversely, and detailed studies report mode conversion, multi-timescale coupling, and stability trade-offs when virtual inertia is added to grid-following generation [2]–[5].

These observations suggest a question that is different from the usual “how much inertia is enough?”:

How much inertia, placed where, is required to keep network-dominated modes spectrally separated from controller-dominated modes?

The distinction matters because a controller/network encounter need not be visible in a scalar connectivity measure. Generalized-Fiedler metrics now combine topology and nodal inertia to characterize dynamic connectivity [6], while complex-frequency theory provides a geometric interpretation of linear modes [7]. Those developments are complementary to the present work but do not answer the sector-level time-scale separation problem considered here.

The central thesis of this paper is

$$\boxed{\text{inertia is both stored kinetic energy and a spectral-separation resource.}} \quad (1)$$

The proposed framework has four contributions.

- C1. Feedback-sector separation.** A controller-aware closed-loop model is partitioned into network and controller subspaces inside algebraically meaningful feedback sectors. A Sylvester separation margin quantifies whether those subspaces can remain weakly mixed.
- C2. Inertia-resonance theory.** We derive the $\eta^{-1/2}$ migration law under uniform inertia scaling, a closed-form controller-resonant inertia threshold, and a two-mode hybridization formula that explains mode exchange and loss of model separation.
- C3. Inertia geometry and steering.** We define a signed inertia-steering field and show that equal additions of inertia at different buses can move the same critical pole in opposite directions. Equal-total-inertia placements can therefore be stable or unstable solely because the budget is spatially redistributed.
- C4. Inertia/controller co-design.** Dense phase diagrams and first-crossing calculations quantify the intervention re-

TABLE I
QUANTITATIVE EVIDENCE USED IN THE IEEE 39-BUS STRESS STUDY

Experiment	Scale / result
(η, K_p) co-design phase map	2,232 points
Equal-total-inertia placements	1,200
Stable / unstable equal-budget cases	316 / 884
Critical-abcissa range, equal budget	$[-0.326, +1.151] \text{ s}^{-1}$
Bus-31 local-inertia sweep	250 points
Re-entrant crossings	0.1507, 1.5169 p.u. of H_{31}^{nom}
Bus/operating-point inertia steering map	162 interventions
Minimum finite inertia crossings at $\eta = 0.25$	0.826–55.57
PLL retuning threshold at $\eta = 0.25$	$K_p = 1.0486$

quired at each bus and compare it with controller detuning. The resulting remediation map identifies locations where added inertia is efficient, ineffective, or initially destabilizing.

The paper does *not* claim that modal proximity itself is new. Open-loop modal proximity and multi-timescale interaction are active topics in converter stability [4]. The contribution pursued here is a network-level, feedback-sector formulation that makes inertia placement and controller retuning comparable through the same spectral-separation object.

II. CONTROLLER-AWARE FEEDBACK SECTORS

A. Reduced closed-loop operator

Consider a linearization of a nonlinear AC-DAE around an operating point. Retain a set of terminal/network states and eliminate internal controller states by a Schur complement. The reduced nonlinear eigenvalue problem can be written

$$F(s)x = 0, \quad F(s) = s^2M + sD + K + \Pi(s), \quad (2)$$

where K is the tangent electrical stiffness and $\Pi(s)$ is the frequency-dependent self-energy induced by eliminated controller/filter states. The poles of $\Pi(s)$ therefore retain controller memory even when the retained model is compact.

The cycle-algebra viewpoint developed in the companion scalarizability framework motivates a decomposition of the retained port space into invariant or approximately invariant closed-loop feedback sectors

$$\mathcal{V} = \mathcal{V}_1 \oplus \cdots \oplus \mathcal{V}_R. \quad (3)$$

Within one sector, choose coordinates that separate network-dominated and controller-dominated states:

$$A_r = \begin{bmatrix} A_{g,r}(M) & B_r \\ C_r & A_{c,r} \end{bmatrix}. \quad (4)$$

The sector formulation is important: a network can be globally high order while the dangerous controller/network encounter lives in a low-dimensional irreducible feedback block.

B. Sylvester separation

For square matrices A and B , define the classical separation

$$\text{sep}(A, B) = \sigma_{\min}(I \otimes A - B^T \otimes I). \quad (5)$$

This quantity controls the conditioning of the Sylvester equation $AX - XB = C$ [8]. To compare separation with the strength of the network/controller coupling, define

$$\mathfrak{S}_r = \frac{\text{sep}_G(A_{g,r}, A_{c,r})}{\sqrt{\|B_r\|_G \|C_r\|_G}} \quad (6)$$

using a physically meaningful metric G when non-unitary coordinate changes must be allowed.

Theorem 1 (Sector mixing bound). *Assume $\text{sep}(A_g, A_c) > 0$. For a first-order block-diagonalizing transformation with graph matrix X satisfying*

$$A_g X - X A_c = -B, \quad (7)$$

one has

$$\|X\|_F \leq \frac{\|B\|_F}{\text{sep}(A_g, A_c)}. \quad (8)$$

The analogous controller-to-grid correction satisfies $\|Y\|_F \leq \|C\|_F / \text{sep}(A_c, A_g)$.

Proof. Vectorization gives

$$(I \otimes A_g - A_c^T \otimes I) \text{vec}(X) = -\text{vec}(B). \quad (9)$$

Taking the smallest singular value of the Sylvester operator yields (8). The second inequality is identical with the roles reversed. $\square$

Remark 1. A small $\mathfrak{S}_r$ is not itself a stability certificate. It states that an invariant-subspace separation used by reduced or decoupled models is ill-conditioned relative to the coupling. Stability must still be computed from the full sector.

III. INERTIA AS A SPECTRAL SEPARATOR

A. Uniform inertia migration

Consider first the conservative sector eigenproblem

$$K_r v_r = \omega_r^2 M_r v_r. \quad (10)$$

Uniformly scale the inertia inside the sector as

$$M_r(\eta) = \eta M_r^0, \quad \eta > 0. \quad (11)$$

Theorem 2 (Inertia migration law). *If ω_{r0} is a nonzero natural frequency at $\eta = 1$, then*

$$\omega_r(\eta) = \frac{\omega_{r0}}{\sqrt{\eta}}. \quad (12)$$

Proof. Substitution into (10) gives $K_r v_r = \omega_r^2 \eta M_r^0 v_r$. Comparing with the $\eta = 1$ generalized eigenproblem gives $\omega_r^2 = \omega_{r0}^2 / \eta$. $\square$

Thus low inertia sweeps network modes upward through the controller spectrum. If a controller pole is

$$\mu_\ell = -\sigma_\ell + j\omega_{c,\ell}, \quad (13)$$

the simplest frequency-encounter threshold is

$$\eta_{r\ell}^* = \left(\frac{\omega_{r0}}{\omega_{c,\ell}} \right)^2. \quad (14)$$

Equation (14) is not a stability boundary by itself; it identifies the inertia level at which detuning vanishes in the conservative approximation.

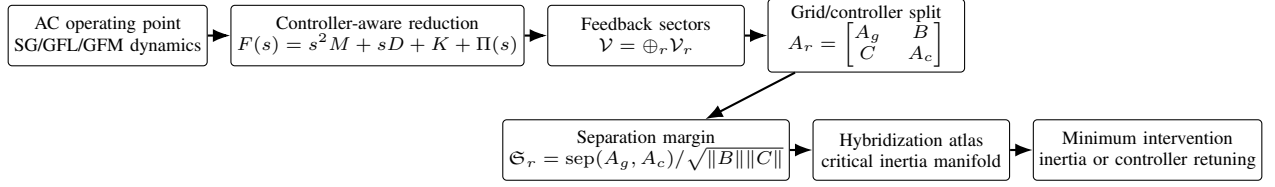


Fig. 1. Proposed low-inertia analysis. The key object is not total inertia but spectral separation between network and controller dynamics inside the relevant closed-loop feedback sectors.

B. Distributed inertia-resonance manifold

For heterogeneous inertia $M(\mathbf{m}) = \text{diag}(m_1, \dots, m_n)$, project onto sector r and define the controller-resonance manifold

$$\mathcal{R}_{r\ell} = \{\mathbf{m} : \det(K_r - \omega_{c,\ell}^2 M_r(\mathbf{m})) = 0\}. \quad (15)$$

This formulation makes explicit why the *location* of inertia matters: two inertia vectors with the same sum need not have the same projection onto the critical sector.

C. Two-mode hybridization

Near an isolated encounter, project the full problem onto one network mode and one controller mode. A minimal local model is

$$\begin{bmatrix} s - s_g(\eta) & -g \\ -h & s - \mu_c \end{bmatrix} z = 0. \quad (16)$$

Proposition 1 (Hybridized poles). *The two coupled poles are*

$$s_{\pm} = \frac{s_g + \mu_c}{2} \pm \sqrt{\left(\frac{s_g - \mu_c}{2}\right)^2 + gh}. \quad (17)$$

Proof. Set the determinant of (16) to zero and apply the quadratic formula. $\square$

Far from the encounter, $|s_g - \mu_c| \gg |gh|^{1/2}$ and the modes retain their grid/controller identities. Near the encounter, the square-root term is dominated by coupling and mode identity is exchanged. This is the mechanism behind the use of the term *hybridization*; the concept is related to modal-proximity analyses in converter systems, but here it is embedded into a network-level inertia-placement problem.

D. Inertia and damping are different steering directions

For the nonlinear eigenvalue problem (2), let x_k, y_k be right/left eigenvectors and

$$T_k = y_k^* F_s(\lambda_k) x_k. \quad (18)$$

For local nodal perturbations m_i, d_i ,

$$\frac{\partial \lambda_k}{\partial m_i} = -\frac{\lambda_k^2 y_k^* E_i x_k}{T_k}, \quad (19)$$

$$\frac{\partial \lambda_k}{\partial d_i} = -\frac{\lambda_k y_k^* E_i x_k}{T_k}. \quad (20)$$

Hence

$$\frac{\partial \lambda_k}{\partial m_i} = \lambda_k \frac{\partial \lambda_k}{\partial d_i}. \quad (21)$$

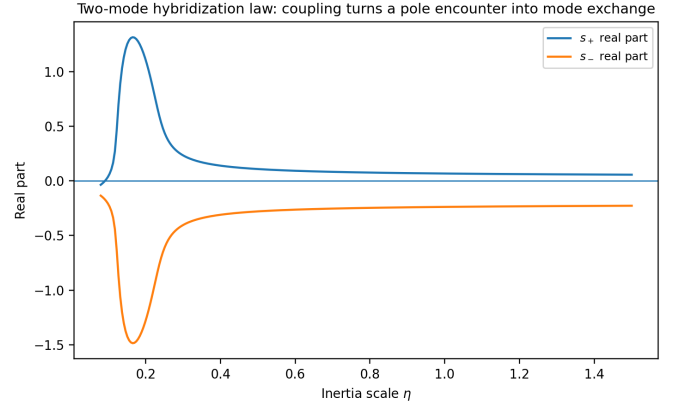


Fig. 2. Illustrative two-mode hybridization law (17). Coupling converts a grid/controller pole encounter into a mode exchange; the figure is conceptual, not IEEE 39-bus data.

For an oscillatory mode, virtual inertia therefore rotates/scales the local damping sensitivity in the complex plane rather than duplicating it. This motivates two-parameter controller co-design instead of a single “add damping” rule.

E. Signed inertia steering field

For a selected critical mode, define the local inertia-steering vector

$$\mathbf{s}_{i,k}^M = \begin{bmatrix} \partial \Re \lambda_k / \partial M_i \\ \partial \Im \lambda_k / \partial M_i \end{bmatrix}. \quad (22)$$

Its first component is a signed local stability authority: $-\partial \Re \lambda_k / \partial M_i > 0$ means that a small local inertia increase moves the mode left, while the opposite sign means that the same positive inertia addition moves the mode right. Equation (21) shows why no universal sign follows from $M_i > 0$ in a non-normal controller-rich system: left/right modal geometry and the complex transversality denominator T_k enter the sensitivity. The field (22) therefore provides a location-dependent complement to aggregate inertia metrics.

IV. PLANNING AND CO-DESIGN IMPLICATIONS

A. Minimum separation inertia

A planning formulation can distinguish frequency-security requirements from spectral-separation requirements. Let $C_M(\mathbf{m})$

be an inertia-placement cost. Define

$$\begin{aligned} \min_{\mathbf{m}} \quad & C_M(\mathbf{m}) \\ \text{s.t.} \quad & \mathfrak{S}_r(\mathbf{m}, \boldsymbol{\kappa}) \geq \mathfrak{S}_{\min}, \quad r \in \mathcal{R}_c, \\ & \mathbf{m}_{\min} \leq \mathbf{m} \leq \mathbf{m}_{\max}, \end{aligned} \quad (23)$$

where $\mathcal{R}_c$ is a set of critical feedback sectors. Equation (23) defines a *spectral-separation inertia reserve*; it is different from an aggregate RoCoF constraint.

B. Inertia/controller detuning co-design

Because both network and controller poles are movable, a cheaper formulation is

$$\begin{aligned} \min_{\Delta \mathbf{m}, \Delta \boldsymbol{\kappa}} \quad & C_M(\Delta \mathbf{m}) + C_\kappa(\Delta \boldsymbol{\kappa}) \\ \text{s.t.} \quad & \text{sep}_G(A_{g,r}(\mathbf{m} + \Delta \mathbf{m}), A_{c,r}(\boldsymbol{\kappa} + \Delta \boldsymbol{\kappa})) \\ & \geq s_{\min}, \quad r \in \mathcal{R}_c. \end{aligned} \quad (24)$$

The optimizer can therefore choose between adding virtual inertia, changing controller bandwidth, changing PLL gains, or combining small changes. The industrial implication is direct: a low-inertia warning need not automatically imply a large BESS installation if a small controller detuning produces greater separation.

V. IEEE 39-BUS FALSIFICATION STUDY

A. Model and experimental design

The numerical study uses the solved IEEE 39-bus AC operating point and the AC-consistent reduced SG/GFL network model developed for the companion cycle-dynamics study. Ten generation ports are retained after Schur elimination of load buses. Six ports use classical swing/source dynamics and four ports (buses 30, 34, 37, and 38) use GFL models with PI PLLs and first-order P/Q current-control surrogates. The placement mirrors a common low-inertia conversion pattern in IEEE 39-bus studies [10]. The study is deliberately falsification-oriented: it asks whether total inertia, monotonic inertia addition, and inertia-only remediation survive a coupled controller/network stress test. Unless otherwise stated, the PLL stress family uses $K_i = 400$ and $K_p = 1$. The model is reduced and is not presented as a substitute for EMT/full-order converter validation.

B. A dense low-inertia phase diagram

The first campaign uniformly scales synchronous inertia by η while sweeping K_p . The 2,232-point map in Fig. 3 overlays the full closed-loop stability boundary with grid/controller pole proximity and the Sylvester separation margin. Low $\mathfrak{S}$ forms narrow bands at low inertia, and the stability boundary bends around the same controller/network interaction region rather than following a monotone inertia threshold. For $K_p = 1$, the uniform-inertia path crosses the small-signal boundary near $\eta \approx 0.309$. The co-design frontier extracted from the same data is also non-monotone: the sampled minimum K_p for stability ranges from 0.96 to 1.36, with the largest requirement around $\eta = 0.18$.

C. Equal total inertia does not order stability

The effective synchronous-inertia budget is next fixed at 30% of nominal and randomly redistributed among the six synchronous generation buses. The budget is identical in all 1,200 trials. Nevertheless, the critical spectral abscissa ranges from -0.326 to $+1.151 \text{ s}^{-1}$, with 316 stable and 884 unstable placements. Fig. 4 compresses four views of the campaign. The placement cloud occupies a continuous low-dimensional geometry even though its total inertia is constant. Stable placements assign on average 22.2% of the budget to bus 39, compared with 8.25% for unstable placements. Conversely, unstable cases allocate more budget on average to buses 31, 32, 35, and 36. The distribution of the budget also matters: the normalized placement entropy is 0.782 on average for stable cases and 0.734 for unstable cases. In the highest entropy quintile, 37.5% of placements are stable, compared with only 15.4% in the lowest quintile. This trend is case-specific, but it reinforces the main nonordering result

$$\sum_i M_i = \text{constant} \not\Rightarrow \alpha(A_{cl}) = \text{constant}. \quad (25)$$

D. Local inertia can create a re-entrant instability window

A stronger test fixes every synchronous machine except bus 31 at 15% of nominal inertia and varies only H_{31} . The system is stable at very small bus-31 inertia, becomes unstable, and then restabilizes. The crossings are

$$H_{31}/H_{31}^{\text{nom}} \approx 0.1507 \quad \text{and} \quad 1.5169. \quad (26)$$

The maximum critical real part reaches approximately $+0.251 \text{ s}^{-1}$ near $H_{31}/H_{31}^{\text{nom}} = 0.205$. Fig. 5 shows why this is a hybridization rather than a monotone energy effect: the decoupled grid pole is swept through the controller band, the full coupled mode follows its own hybrid trajectory, and the subspace separation becomes small in the encounter region. The result is not a universal statement that inertia is destabilizing; it is a counterexample to the stronger monotonic claim that adding inertia at an arbitrary location must improve the coupled small-signal margin.

E. The inertia steering field identifies useful and harmful locations

At a uniform inertia scale $\eta = 0.25$, the baseline stress case has $\Re \lambda_c = +0.0175 \text{ s}^{-1}$. Identical increments of the model's effective inertia coefficient are then applied one bus at a time. The signed steering field in Fig. 6(a)–(b) shows opposite local directions: the same increment moves the critical pole left at buses 33, 39, and 31, but right at buses 32, 35, and 36. For the common increment used in panel (b), bus 33 changes $\Re \lambda_c$ by -0.0361 s^{-1} and stabilizes the case, whereas bus 35 changes it by $+0.0282 \text{ s}^{-1}$ and worsens the instability.

The finite first-crossing calculation strengthens this contrast. Added effective inertia of only 0.826 units at bus 33 reaches the stability boundary, compared with 7.293 at bus 39 and 55.57 at bus 35. Buses 31, 32, and 36 do not reach a stability crossing within a search range exceeding 593 units. Thus the

Low-inertia phase diagram: stability is organized by controller/network separation

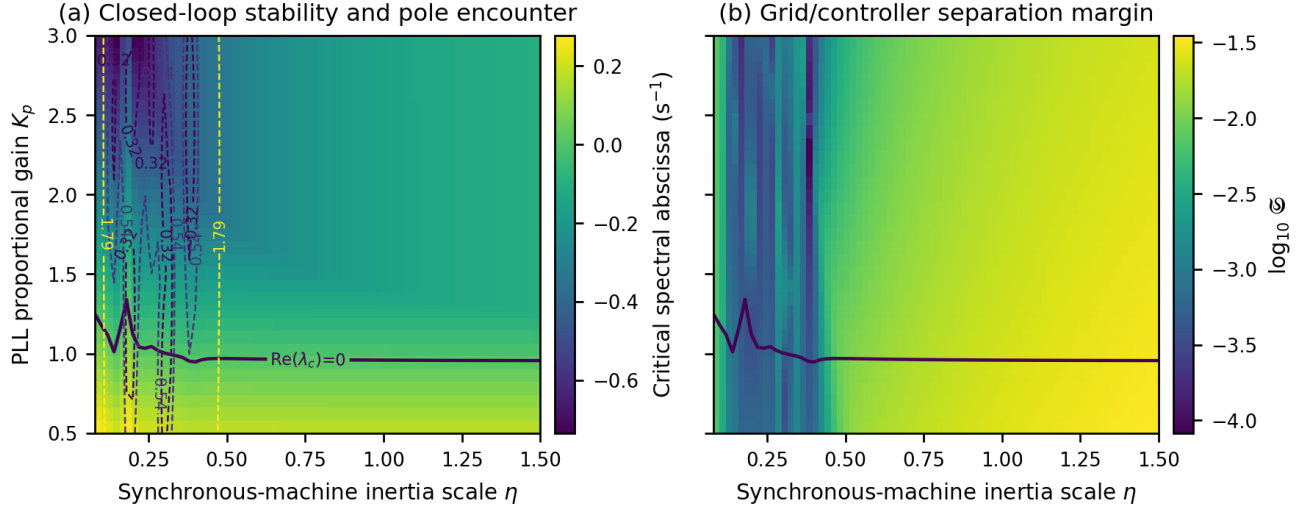


Fig. 3. Dense inertia/controller phase diagram. (a) Full coupled critical spectral abscissa with the $\Re\lambda_c = 0$ boundary and contours of nearest decoupled grid/controller pole distance. (b) The corresponding grid/controller Sylvester separation margin. The controller/network encounter is a structured region in (η, K_p) space rather than a single aggregate-inertia threshold.

finite crossing requirements differ by a factor of 67 among buses that do cross and by a lower bound exceeding $700\times$ when the non-crossing locations are included. The ranking is also operating-point dependent: panel (a) shows sign reversals of $\partial\Re\lambda_c/\partial M_i$ as the base inertia scale changes.

F. Controller detuning can dominate inertia restoration

The same $\eta = 0.25$ stress case is stabilized by changing only the PLL proportional gain. The exact threshold is

$$K_p^* \approx 1.0486, \quad (27)$$

only 4.86% above the stress value $K_p = 1$. This comparison is not an economic cost equivalence and does not imply that PLL retuning replaces physical inertia for RoCoF or nadir requirements. It demonstrates a narrower engineering point: once the limiting mechanism is controller/network hybridization, the cheapest *small-signal* remediation may be controller detuning rather than restoring aggregate inertia. Equations (23)–(24) therefore provide a natural hierarchy: enforce frequency-security inertia first, then use sector separation and signed steering to decide whether additional inertia, a different inertia location, or controller retuning is the efficient stability action.

Total inertia is invariant; stability is not

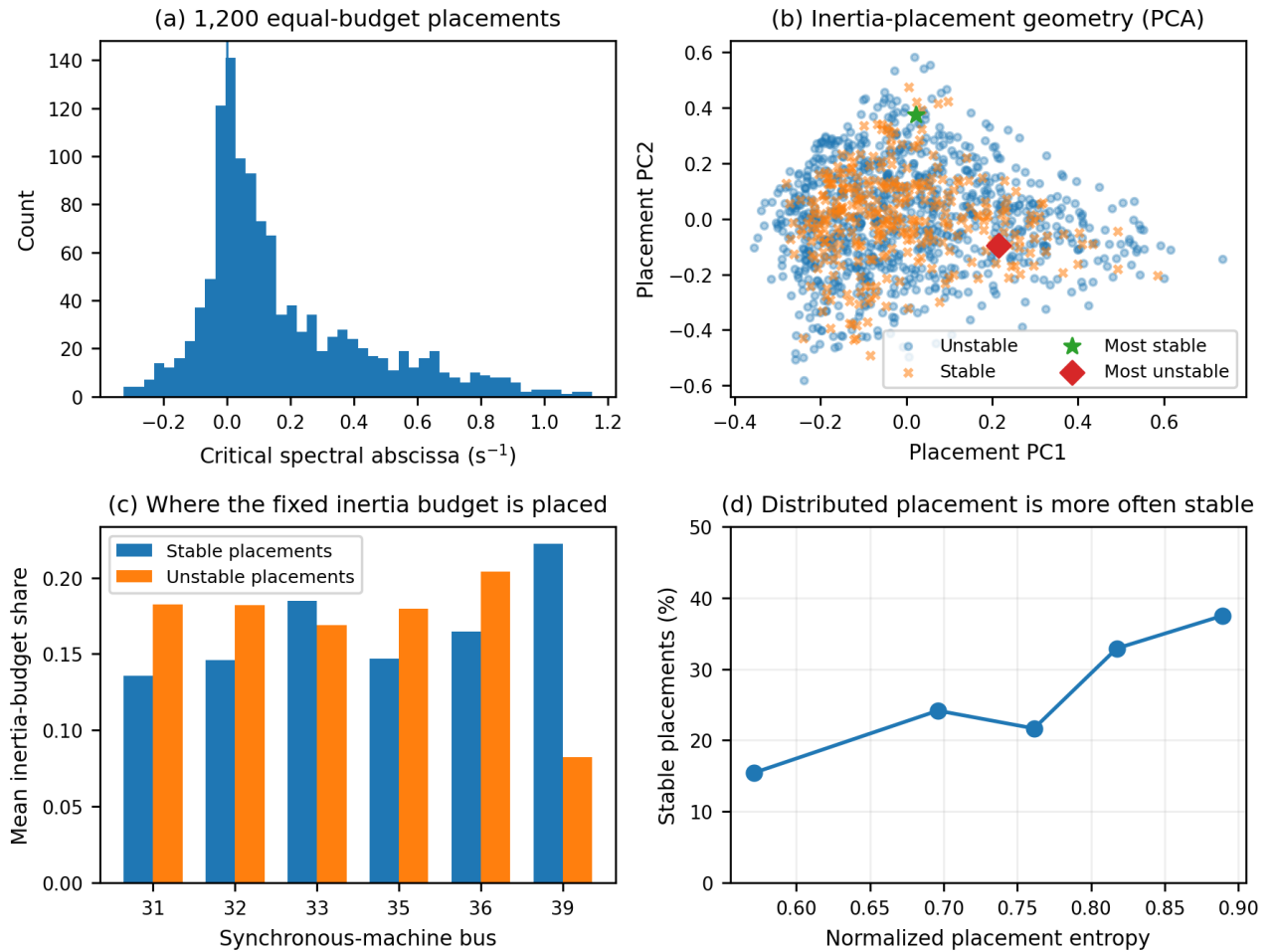


Fig. 4. Equal-total-inertia geometry. (a) Critical-abscissa distribution for 1,200 placements. (b) PCA projection of the six-dimensional fixed-budget placement simplex; stable and unstable outcomes interpenetrate. (c) Mean inertia-budget shares for stable and unstable subsets. (d) Stable fraction versus normalized placement entropy quintile. All cases have the same total synchronous inertia.

Local inertia acts as a detuning coordinate, not a monotone stability resource

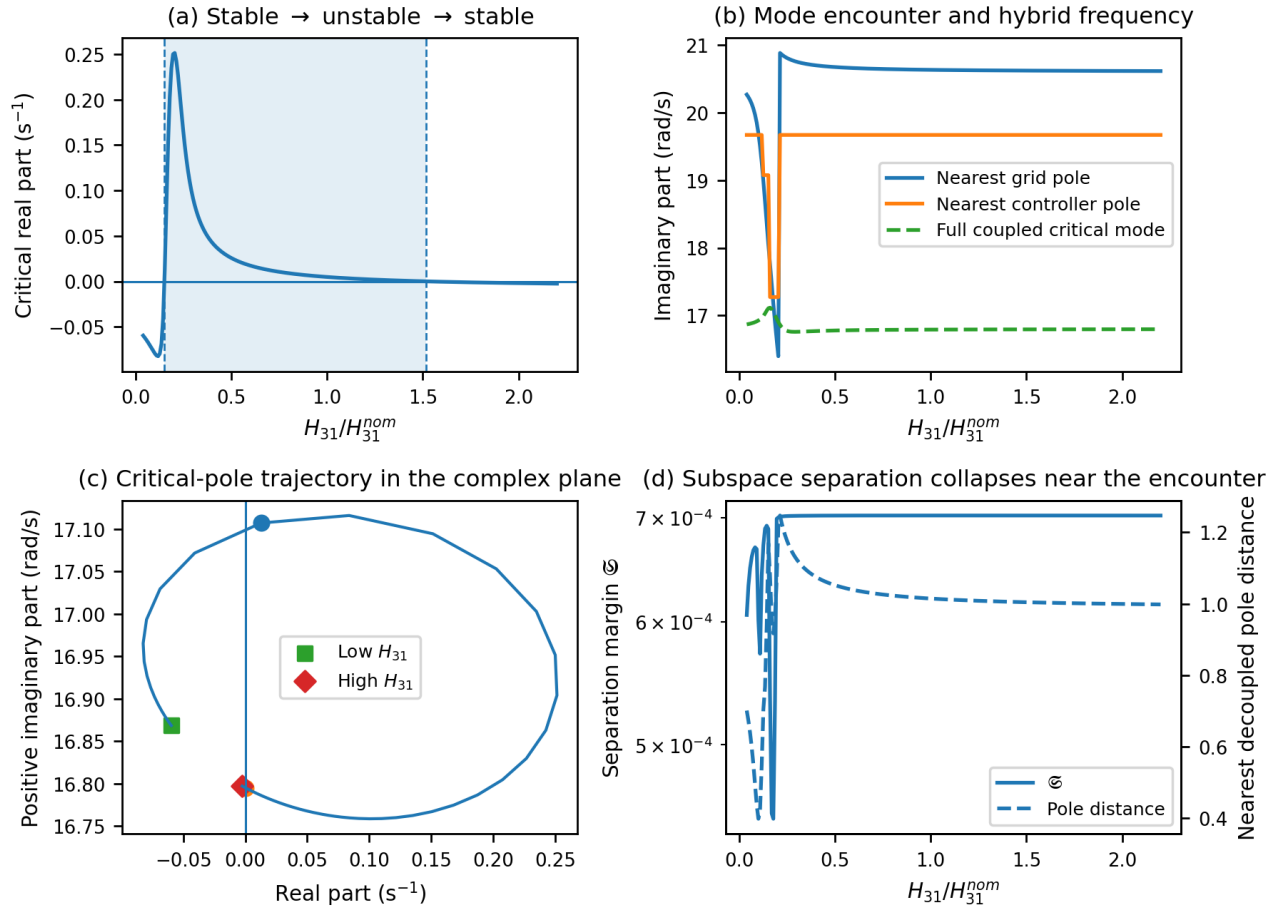


Fig. 5. Re-entrant local-inertia experiment. (a) Stable–unstable–stable spectral abscissa. (b) Nearest decoupled grid/controller pole frequencies and full coupled critical-mode frequency. (c) Critical-pole path in the complex plane. (d) Sylvester separation margin and nearest decoupled pole distance. Inertia acts as a detuning coordinate that can first move a grid mode toward a controller encounter and later away from it.

Where inertia is added matters as much as how much is added

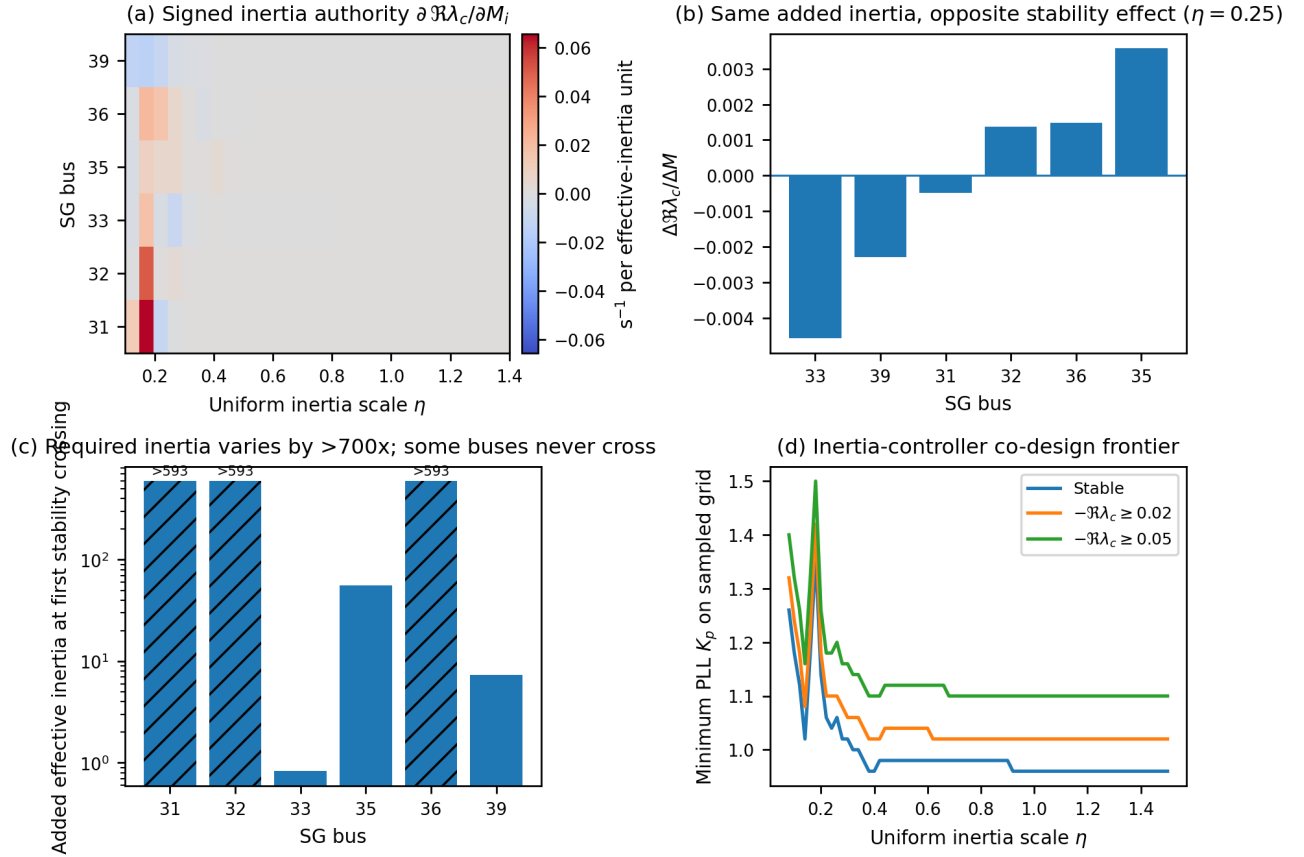


Fig. 6. Inertia-placement authority and remediation. (a) Signed local inertia authority across buses and base inertia levels. (b) Equal positive inertia increments at $\eta = 0.25$ move the critical mode in opposite directions depending on location. (c) Minimum added effective inertia needed for a first stability crossing; hatched bars denote no crossing within the tested > 593-unit range. (d) Minimum sampled PLL K_p required for stability or prescribed small-signal margins as a function of inertia scale.

VI. RELATION TO EXISTING LOW-INERTIA METRICS

The proposed separation view is complementary to, not a replacement for, established metrics.

- **RoCoF and nadir** quantify frequency-security consequences of low kinetic energy. They remain necessary operational constraints.
- **Fiedler/GFV** quantify topology/inertia connectivity and frequency-strength structure [6]; they do not explicitly encode the controller pole spectrum.
- **Modal proximity and impedance methods** capture controller/network interaction at device or system level [2], [4]; the present formulation emphasizes sector-level separation and inertia placement.
- **Model-order reduction** addresses the numerical stiffness produced by overlapping time scales [1]. Here, the same loss of separation is treated as a planning/control phenomenon.

A practical workflow is therefore hierarchical: (i) enforce frequency-security constraints on RoCoF/nadir and physical energy headroom; (ii) identify the feedback sectors that support

controller/network interactions; (iii) use $\mathfrak{S}_r$ and the steering field (22) to determine whether the risk is a spectral-encounter problem; and (iv) select the least intrusive combination of inertia placement and controller detuning. The paper therefore does not propose replacing conventional inertia metrics with a single new scalar. It separates two engineering questions that aggregate inertia merges: *how much inertial energy is available?* and *where does that inertia place the closed-loop spectrum relative to controller dynamics?*

VII. OPEN THEORETICAL PROBLEMS AND FULL-ORDER VALIDATION

Several results remain to be established before the framework can support operational recommendations.

A. Exact feedback-sector extraction

The present IEEE 39-bus experiment uses the coarse grid/controller partition of the reduced state matrix. The next step is to compute the finest simultaneous invariant decomposition of the cycle-generated feedback algebra and evaluate $\mathfrak{S}_r$ only inside the irreducible sectors. This should sharpen the weak correlation observed when a global grid/controller separation is used indiscriminately.

B. Robust separation

Controller poles and inertia vary with operating point. A robust metric is

$$\mathfrak{S}_r^{\text{rob}} = \min_{p \in \mathcal{P}} \mathfrak{S}_r(p), \quad (28)$$

where $\mathcal{P}$ includes load, dispatch, controller uncertainty, topology, and IBR availability. The corresponding co-design problem can enforce separation over a family of operating points.

C. Full-order dynamic models

The strongest remaining validation gate is replication with detailed SG/GFL/GFM and converter inner-loop dynamics. The EPFL IEEE 39-bus benchmark provides low- and very-low-inertia configurations with inverter-connected generation [10]. Full-order Simulink/EMT or ANDES-type validation should test whether the re-entrant inertia window and equal-total-inertia placement reversal persist when current limits, DC-link dynamics, filters, and detailed converter controls are retained.

VIII. CONCLUSION

Low inertia changes more than the kinetic energy available to oppose frequency acceleration. In a controller-rich network it also moves network modes relative to controller poles, changes invariant-subspace separation, and changes the direction in which a local inertia intervention moves the critical closed-loop pole. The paper formalized this role through a Sylvester separation margin, a sector mixing bound, an inertia-resonance manifold, a two-mode hybridization law, and a signed inertia-steering field. The IEEE 39-bus reduced stress study then provided four quantitative counterexamples to inertia-only reasoning: 1,200 placements with identical total inertia produced both stable and unstable cases over a 1.48 s^{-1} abscissa span;

a single-bus inertia sweep generated a stable–unstable–stable window; identical positive inertia additions at different buses moved the critical pole in opposite directions; and the minimum inertia required for a first stability crossing varied from 0.826 to 55.57 effective units while several locations never crossed within a > 593 -unit search range. A 4.86% PLL-gain retuning stabilized the same 25%-inertia stress case without restoring the missing inertia.

These results support a planning principle that is deliberately narrower than “more inertia is bad” or “controller tuning replaces inertia”:

$$\boxed{\text{low-inertia adequacy is jointly an energy, placement, and spectral-separation problem.}} \quad (29)$$

The principal remaining gate is full-order validation with detailed converter inner loops, current limits, DC-link dynamics, and GFM/GFL controls. If the placement reversal and re-entrant window persist there, spectral-separation constraints should become a natural complement to RoCoF/nadir constraints in virtual-inertia and controller co-design.

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